Power Supplies and Connectors: Assignment Answers

SMPS, UPS and Power Connectors

Question 1

List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

Power supply	Device	Why the device needs it
SMPS	Desktop PC	Converts mains AC into the regulated DC voltages needed by the motherboard, drives and graphics card.
	Laptop or mobile phone charger	Provides a compact, lightweight adapter that converts mains AC into the low-voltage DC needed by the battery.
	LED TV or set-top box	Supplies steady DC to the internal circuits from the mains in a small, efficient unit.
UPS	Desktop PC (home or college lab)	Gives battery backup during a power cut so that work can be saved and the system shut down safely.
	Wi-Fi router or network server	Keeps the network equipment running through short power failures and voltage dips.

Question 2

Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections. (Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.)

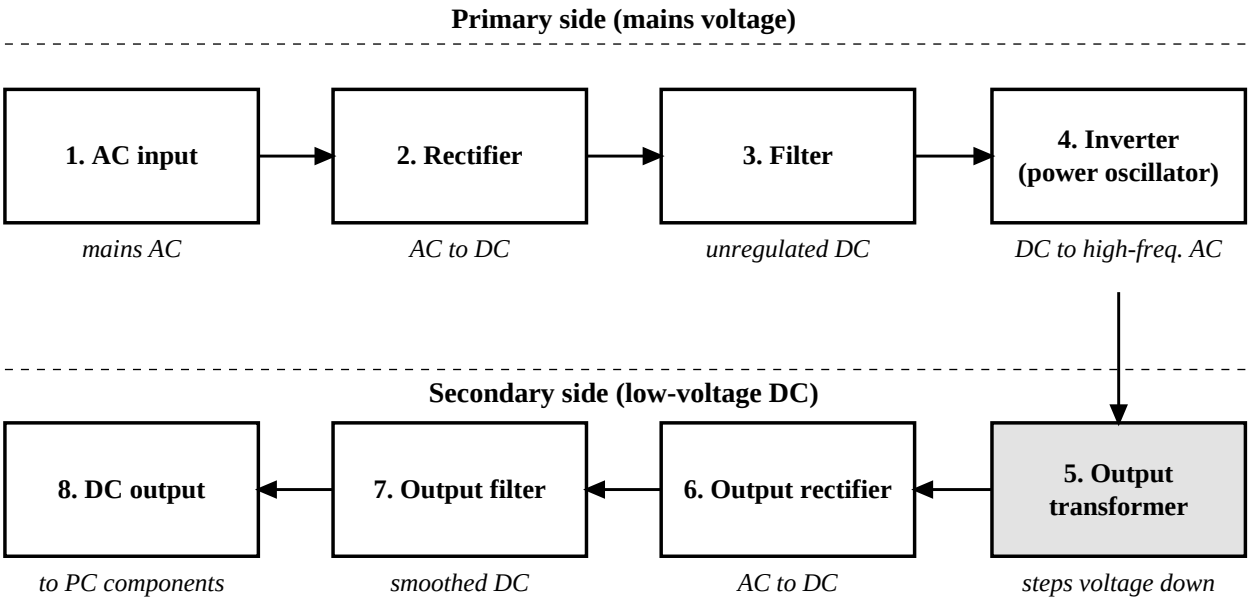
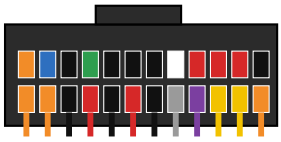
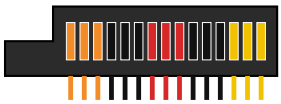
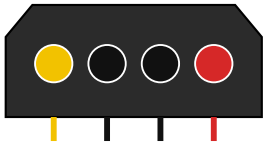
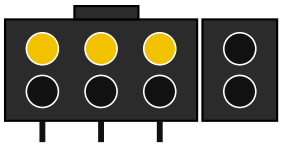


Figure 1: Block diagram of a typical SMPS unit.

No.	Section	Function
1	AC input	Takes the mains AC supply into the SMPS.
2	Rectifier	Converts the AC voltage into DC.
3	Filter	Smooths the rectified output, which is unregulated DC voltage.
4	Inverter	Converts the DC back to high-frequency AC with the help of a power oscillator.
5	Output transformer	Steps the AC voltage down to the required output level.
6, 7	Output rectifier and filter	Rectifies the transformer output (silicon or Schottky diodes) and smooths it using a filter.
8	DC output	Delivers the regulated DC voltages to the motherboard and drives.

Question 3

Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

Image	Name	Devices powered	Safety tip
 *24 pins (2 rows of 12)*	**ATX 24-pin motherboard connector**	The motherboard, and through it the RAM, chipset and expansion slots.	Switch off the SMPS and unplug the mains first; push straight in until the latch clicks and press the latch when removing.
 *15 pins, L-shaped key*	**SATA power connector**	SATA hard disks, SSDs and optical drives.	Align the L-shaped key before pushing in; pull on the plug body, never on the wires.
 *4 pins, two chamfered corners*	**Molex connector (4-pin)**	Older IDE drives, case fans and other accessories.	It fits tightly: rock it gently side to side to remove, and never force it or bend the pins.
 *6-pin + 2-pin = 8-pin*	**PCIe 6/8-pin connector**	Graphics cards that need extra power beyond the PCIe slot.	Do not use a CPU 8-pin cable in its place, as the wiring is different; push in until it clicks.

Note: the images above are simplified colour illustrations of each connector. Wire colours: yellow = +12 V, red = +5 V, orange = +3.3 V, black = ground.

Question 4

Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

1. **Prepare.** Switch off the SMPS and keep the mains cord unplugged. Wear an anti-static wrist strap and use a magnetic screwdriver for the screws.
2. **Install the SMPS** in the case first, with its fan facing the vent, and fix it with screws. Choose a unit with enough wattage for the graphics card.
3. **Motherboard:** fit the motherboard, then connect the **ATX 24-pin** connector to the large socket on the board until the latch clicks.
4. **Processor:** connect the **4+4-pin (8-pin) CPU power** connector to the processor power socket near the processor.
5. **Graphics card:** fit the card in the PCIe x16 slot, then connect the **PCIe 6+2-pin** power cables from the SMPS (one cable for each socket). Newer cards may use a 16-pin 12V-2x6 cable.
6. **Storage devices:** connect a **SATA power** connector from the SMPS to each SATA hard disk or SSD, and a separate **SATA data cable** to the motherboard. An NVMe M.2 SSD needs no cable from the SMPS.
7. **Other devices:** use a **Molex** connector for older drives or fans if required, then connect the monitor and the power cord.
8. **Check and test:** make sure every connector is fully seated and no cable touches a fan, plug in the mains cord, switch on the SMPS and press the power button to confirm that the system starts.

Component	Cable or connector from the SMPS
Motherboard	ATX 24-pin
Processor	CPU 4+4-pin (8-pin)
Graphics card	PCIe 6+2-pin (or 16-pin 12V-2x6)
SATA hard disk / SSD	SATA power connector, plus SATA data cable
NVMe M.2 SSD	None (powered by the motherboard)
Older drives and fans	Molex connector